

Customer Shopping Behavior Analysis

End-to-End Data Analytics Portfolio Project

Tools	Python(Pandas), PostgreSQL, SQL, Power BI
Dataset	3,900 customer purchase records
Workflow	Data preparation → SQL analysis → Dashboard → Business recommendations

1. PROJECT OVERVIEW

This project analyzes customer shopping behavior using transactional data containing 3,900 purchase records. The purpose is to understand spending patterns, customer segments, product preferences, discount usage, shipping behavior, subscription status, and revenue contribution across age groups.

The project follows a practical analytics workflow: raw data is explored and cleaned with Python, transformed data is loaded into PostgreSQL, SQL is used to answer business questions, and Power BI is used to present the results through an interactive dashboard.

Business question: How can a retail company use consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

2. BUSINESS PROBLEM

A retail company wants to better understand its customers' shopping behavior to improve sales, customer satisfaction, and long-term loyalty. The analysis focuses on purchasing patterns across demographics and product categories, together with factors such as discounts, reviews, seasons, payment preferences, subscriptions, and repeat purchases.

The analysis was organized around three practical areas: data preparation and modeling, business-oriented SQL analysis, and visualization of key patterns in Power BI.

Project deliverables

- Clean and transform the raw customer shopping dataset using Python.
- Load the prepared data into PostgreSQL for structured SQL analysis.
- Answer business questions using SQL queries.
- Build an interactive Power BI dashboard.
- Document findings and practical business recommendations.

3. DATASET AND DATA PREPARATION

The original dataset contains 3,900 rows and 18 columns. It includes customer demographics, purchase details, product category information, review ratings, subscription status, shipping type, discount usage, previous purchases, payment method, and purchase frequency.

Area	Examples
Customer information	Age, Gender, Location, Subscription Status
Purchase information	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping behavior	Discount Applied, Previous Purchases, Frequency of Purchases
Transaction experience	Review Rating, Shipping Type, Payment Method

Cleaning and transformation steps

- Checked the dataset structure and summary statistics using Pandas.
- Identified 37 missing values in the Review Rating column.
- Filled missing Review Rating values using the median rating within the corresponding product category.
- Standardized column names to snake_case for easier analysis.
- Created age_group by dividing customer ages into four groups.
- Created purchase_frequency_days by mapping purchase-frequency labels to approximate day intervals.
- Checked discount_applied and promo_code_used for redundancy and removed promo_code_used.
- Connected the prepared DataFrame to PostgreSQL for SQL analysis.

After transformation, the working dataset contains 19 columns: the original analytical fields plus age_group and purchase_frequency_days, with promo_code_used removed.

4. SQL BUSINESS ANALYSIS

PostgreSQL was used to answer business questions from the prepared customer table. The SQL analysis covered revenue, discount behavior, product ratings, shipping, subscriptions, customer segmentation, repeat buying, product popularity, and age-group revenue.

Sr. No.	Business question	Purpose
1	Revenue by Gender	Compared total revenue generated by male and female customers.
2	High-Spending Discount Users	Identified customers who used a discount but still spent at or above the average purchase amount.
3	Top 5 Products by Rating	Found the products with the highest average review ratings.
4	Shipping Type Comparison	Compared average purchase amounts for Standard and Express shipping.
5	Subscribers vs. Non-Subscribers	Compared customer counts, average spend, and total revenue by subscription status
6	Discount-Dependent Products	Identified five products with the highest percentage of discounted purchases.
7	Customer Segmentation	Grouped customers into New, Returning, and Loyal segments based on previous purchases.
8	Top 3 Products per Category	Ranked the most purchased products within each product category.
9	Repeat Buyers & Subscriptions	Checked subscription status among customers with more than five previous purchases.
10	Revenue by Age Group	Calculated the revenue contribution of each age group.

5. PROOF OF SQL ANALYSIS

The following screenshots are included as evidence of the SQL work performed in PostgreSQL/pgAdmin. They show actual query outputs from the project rather than only describing the analysis.

Query 2 — High-Spending Discount Users

The query returns customers whose discount_applied value is 'Yes' and whose purchase amount is at least the overall average purchase amount. The captured result contains 839 rows; the screenshot below shows the first visible records.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

Fig. 1. pgAdmin output for SQL Query 2.

Query 7 — Customer Segmentation

Customers were segmented using previous purchase counts. The captured result shows three segments: Loyal, New, and Returning.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Fig. 2. pgAdmin output for SQL Query 7.

Query 5 — Subscribers vs. Non-Subscribers

The query compares total customers, average purchase amount, and total revenue for subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Fig. 3. pgAdmin output for SQL Query 5.

6. POWER BI DASHBOARD

The Power BI dashboard was built to provide a compact visual view of customer behavior. It includes slicers for subscription status, gender, category, and shipping type, together with KPI cards and category/age-group charts.

Dashboard elements visible in the implementation

- Number of customers, average purchase amount, and average review rating.
- Percentage of customers by subscription status.
- Revenue and sales by product category.
- Revenue and sales by age group.
- Interactive filtering through subscription, gender, category, and shipping-type controls.

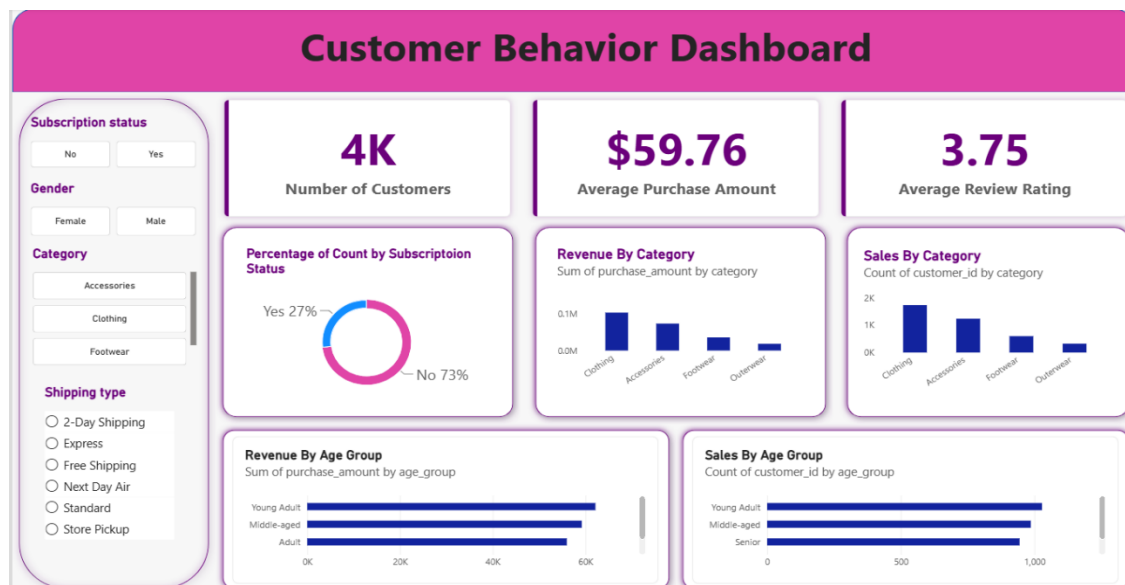


Fig. 4. Power BI Customer Behaviour Dashboard created for the project.

7. KEY FINDINGS

Subscription analysis: The SQL output shows 1,053 customers with subscription status 'Yes' and 2,847 with status 'No'. Average spend is \$59.49 for subscribers and \$59.87 for non-subscribers, while total revenue is \$62,645 and \$170,436, respectively.

Customer segmentation: The captured segmentation result contains 3,116 Loyal customers, 701 Returning customers, and 83 New customers based on previous purchase counts.

Discount behavior: The project specifically analyzes customers who use discounts while still spending above the average, and also identifies products with high proportions of discounted purchases.

Dashboard overview: The Power BI snapshot reports approximately 4K customers, an average purchase amount of \$59.76, and an average review rating of 3.75. The subscription visual shows 73% No and 27% Yes.

Age-group revenue: The analysis calculates revenue contribution across Young Adult, Middle-aged, Adult, and Senior groups, allowing the business to compare customer value across age segments.

8. BUSINESS RECOMMENDATIONS

- Boost subscription engagement: Use the subscription analysis to review the value proposition and consider benefits that encourage suitable customers to subscribe.
- Strengthen customer loyalty: Use the Loyal and Returning segments to design retention and repeat-purchase initiatives.
- Review discount usage: Identify products with high discount rates and evaluate whether discounts are generating incremental demand while protecting margins.
- Support product positioning: Use product rating and purchase-volume analysis to inform merchandising and promotional campaigns.
- Use age-group insights: Tailor campaigns and product communication according to revenue contribution across customer age groups.
- Use dashboard filters for deeper analysis: Combine category, gender, subscription, and shipping filters to investigate specific customer groups.

9. CONCLUSION

This project demonstrates an end-to-end data analytics workflow from raw transactional data to business-facing insights. Python was used for data preparation and feature engineering, PostgreSQL was used for structured business analysis, and Power BI was used to communicate the results visually.

The project also provided practical experience with data cleaning, SQL aggregation and window functions, database integration, dashboard design, and translating analytical results into business recommendations.

Tools & Technologies: Python • Pandas • PostgreSQL • SQL • pgAdmin 4 • Power BI
• Jupyter Notebook

10. PORTFOLIO PROJECT FILES

File	Purpose
Customer_Shopping_Behavior_Analysis.ipynb	Python data loading, exploration, cleaning and feature engineering.
customer_behavior_sql_file.sql	SQL business questions and analysis queries.
customer_behavior_dashboard.pbix	Interactive Power BI dashboard.
customer_shopping_behavior.csv	Source dataset used for analysis, subject to its original usage/redistribution terms.
Customer_Shopping_Behavior_Analysis.pdf	Project documentation, findings, screenshots and recommendations.
README.md	GitHub-facing project overview and documentation.

NOTE ON SOURCE MATERIAL: THE PROJECT WORKFLOW WAS LEARNED THROUGH A GUIDED TUTORIAL. THIS REPORT IS WRITTEN SPECIFICALLY FOR THIS PORTFOLIO REPOSITORY AND DISTINGUISHES THE HANDS-ON ANALYTICAL COMPONENTS FROM THE ORIGINAL LEARNING MATERIAL.